Introductory Physics

Science

May, 2026

Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Introductory Physics (A09888)

Short Title: Introductory Physics
Faculty Science and Computing
Department Science
Cluster Mathematics and Physics
Author CWALSH
Credits 5 **Level:** Introductory

Description of Module / Aims

This module introduces the student to the fundamental principles of classical mechanics, properties of matter and wave motion. The module contains a practical component where the student can develop measurement, logical thinking, analysis and report-writing skills.

Programmes

PHYI-0001	BSc (Common Entry) (WD_SSCIE_D)	1 / 1 / M
PHYI-0001	BSc (Hons) in Agricultural Science (WD_SAGRI_B)	1 / 1 / M
PHYI-0001	BSc (Hons) in Food Science and Innovation (WD_SFSIN_B)	1 / 1 / M
PHYI-0001	BSc (Hons) in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_B)	1 / 1 / M
PHYI-0001	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	1 / 1 / M
PHYI-0001	BSc (Hons) in Physics for Modern Technology (WD_KPHTE_B)	1 / 1 / M
PHYI-0001	BSc (Hons) in Science (Common Entry) (WD_SSCCM_B)	1 / 1 / M
PHYI-0001	BSc in Food Science with Business (WD_SFDSC_D)	1 / 1 / M
PHYI-0001	BSc in Molecular Biology with Biopharmaceutical Science (WD_SMBIO_D)	1 / 1 / M
PHYI-0001	BSc in Pharmaceutical Science (WD_SPHSC_D)	1 / 1 / M

Indicative Content

- Introduction to physics: SI units, prefixes, scientific notation; measurement; vectors and scalars
- Mechanics: motion in one dimension, Newton's laws of motion; work, energy and power; uniform circular motion; conservation laws - mechanical energy, total energy and momentum
- Properties of matter: properties & structure of matter; pressure, elasticity, viscosity and surface tension
- Waves and sound: wave motion and vibrations, wave classification and properties; sound waves, the characteristics of sound and the Doppler effect

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe and explain the fundamental principles of classical mechanics.
2. Describe and explain the fundamental principles of the properties of matter.
3. Describe and explain the fundamental principles of vibrations and wave motion.
4. Complete experiments, record and analyse experimental data and present results in the form of a report.
5. Apply problem-solving skills to generate solutions to problems in mechanics, properties of matter and wave motion.

Learning and Teaching Methods

- Lectures. The lectures will be used to present new topics and their related concepts. Problem solving techniques will also be used in class to analyse physical situations and find numerical solutions.
- Tutorials. On-line tutorials linked to core text.
- Laboratory-based practicals. Integrated practical programme designed to run in parallel with lectures and allow the student to develop a range of experimental skills. The importance of planning of experiments, data acquisition, and analysis will be emphasised.
- Self-directed study.

Assessment Methods

	Weighing		Outcomes Assessed
Final Written Examination	40%		1,2,3,5
Continuous Assessment	60%		
Tutorial/Problem Sheets		20%	5
Practical		40%	4

Assessment Criteria

<40%: Inability to demonstrate a basic understanding of the fundamental concepts of classical mechanics, properties of matter and wave motion. Unable to carry out practical skills in the physics laboratory.

40%-49%: Ability to demonstrate a basic understanding of the fundamental concepts of classical mechanics, properties of matter and wave motion. Ability to provide partial solutions to numerical problems. Has a basic level of competency and skills in the physics laboratory.

50%-59%: Ability to clearly describe and define concepts encountered in classical mechanics, properties of matter and wave motion. Ability to analyse physical situations and apply suitable problem-solving techniques to find numerical solutions. Good practical skills in the physics laboratory.

60%-69%: In addition to the above, demonstrates a very good understanding of classical mechanics, properties of matter and wave motion, as well as the ability to apply problem-solving techniques to new similar problems. Has a high level of practical skills in the physics laboratory. Good clear reporting of practical work.

70%-100%: All the above to an excellent level. Demonstrates an ability to put a numerical solution into a context and assess whether such solutions are meaningful. Has a very high level of practical skills and efficiency in the physics laboratory. Can critically assess experiments undertaken.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Tutorial	8	
Practical	16	
Independent Learning	75	

Essential Material

- Giancoli, D. _Physics:Principles__with Applications, Global edition_. 7th ed. New Jersey: Pearson, 2015.

Supplementary Material

- Cutnell, J. and K. Johnson. _Physics_. 9th ed. United States of America: Wiley, 2012.
- Halliday, D., R. Resnick and J. Walker. _Fundamen__tals of Physics Extended_. 10th ed. United States of America: Wiley, 2013.
- Knight, R., B. Jones and S. Field. _College Physics: A Strategic Approach Technology Update_. 3rd ed. United States of America: Pearson, 2015.
- Walker, J. _Physics Technology Update_. 4th ed. New Jersey: Pearson, 2014.

Requested Resources

- SCIENCE LAB: Physics